Multispectral Image Recompression in Ciphertext Domain With Texture Block Decision and 3D-MDCT

Xiaoran Leng, Weijia Cao, *Member, IEEE*, Tao Yu, and Xingfa Gu

***Abstract*— The rapid advancement of remote sensing (RS) technology has posed increasing demands for secure and efficient processing of multispectral data. However, conventional joint image encryption and compression schemes, originally developed for natural images, are not well suited to the specific requirements of multispectral RS scenarios, such as managing interband redundancy, capturing spatial texture variation, and preserving spectral consistency for downstream applications. To address these challenges, we propose a joint encryption and compression algorithm for multispectral images (JECA-MS), the first joint encryption and compression framework specifically designed for multispectral RS images with support for ciphertext domain recompression. The JECA-MS incorporates four key innovations: 1) an adaptive two-size texture block decision (TBD) strategy that classifies image regions into strong and weak texture blocks (WTBs), reducing data volume in weak-texture areas by up to fourfold; 2) a modified 3-D discrete cosine transform (3D-MDCT) that enhances spatial–spectral decorrelation, particularly in homogeneous regions such as clouds and water; 3) a ciphertext domain recompression mechanism that enables flexible adjustment of compression ratios (CRs) without decryption; and 4) a dedicated JECA-MS coding format (JECA-MS-CF) for efficient data encapsulation and compatibility with RS data structures. Extensive experiments show that the JECA-MS achieves 55% and 36% improvements in CRs for water and cloud images, while reducing encoding and decoding time by 39% and 68%, compared to state-of-the-art methods. Security evaluation shows that the JECA-MS can resist statistical attacks, achieve a tradeoff between lightweight encryption and compression performance. This work offers a flexible solution for secure and efficient RS data management.**

***Index Terms*— Image security, JPEG, multispectral image compression, recompression, remote sensing (RS).**

Received 6 September 2024; revised 25 January 2025 and 25 May 2025; accepted 8 July 2025. Date of publication 16 July 2025; date of current version 31 July 2025. This work was supported in part by Hainan Provincial Natural Science Foundation of China under Grant 624MS126; and in part by the Major Science and Technology Innovation Project of Shandong Province, China, under Project 2025CXGC010113. *(Xiaoran Leng and Weijia Cao contributed equally to this work.) (Corresponding author: Weijia Cao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3589555

I. Introduction

RECENT advancements in remote sensing (RS) technology have dramatically increased the need for secure and efficient management of large volumes of multispectral RS data. With over 2000 satellites currently in orbit serving diverse Earth observation missions, massive datasets are generated daily, posing substantial challenges for the efficiency and security of data storage and transmission [1]. These datasets carry sensitive environmental and strategic information, such as water resources, agricultural patterns, or atmospheric dynamics, which must be protected from unauthorized access during transmission and storage. Conventional image encryption schemes rely on pixel-level encryption, resulting in prohibitive computational demands incompatible with satellite onboard processing capabilities [2]. RS images are characterized by a large amount of redundancy and changing spatial texture features, especially large weak-texture regions show extremely high spatial and spectral correlation. Removing these correlations by compression technology can significantly reduce the computational demands. Consequently, the joint encryption and compression algorithm (JECA) that integrates encryption directly into the compression workflow is imperative for managing large-scale RS data securely and efficiently. Existing JECA typically necessitates ciphertext decryption before data recompression, introducing severe security vulnerabilities such as potential key leakage, especially in relay satellite communication scenarios. Therefore, developing JECA methods capable of ciphertext domain recompression—allowing data to be recompressed securely and flexibly without decryption—has become critical.

RS image compression methods can generally be categorized into lossless and lossy approaches. Early satellites typically employed lossless compression techniques such as differential pulse code modulation (DPCM) [3], adaptive

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DPCM (ADPCM) [4], and run-length encoding (RLE) due to their simplicity and ease of hardware implementation. Nevertheless, the limited compression ratios (CRs) achievable by lossless compression are insufficient to cope with the growth in RS data volume. Lossy compression methods, which significantly enhance compression ratios by selectively discarding less critical high-frequency information, have thus gained popularity. Common lossy compression methods include as following.

1) Compressed sensing (CS)-based methods, exploiting inherent signal sparsity for efficient data reduction during image acquisition [5], [6].

2) Deep neural network (DNN)-based methods, leveraging learned feature representations for improved compression performance [7], [8], [9]. Despite their strengths, these methods remain computationally intensive and exhibit limited generalization capabilities, restricting applicability in RS scenarios.

3) Transformation-based methods, converting spatial data to frequency-domain representations to facilitate efficient compression. Among them, JPEG [10] remains widely adopted due to its balance between implementation simplicity and performance efficiency, exemplified by systems such as SPOT-5 and Ikonos-2 [11]. Recent research has expanded JPEG's capabilities to include encryption and recompression functionalities.

JPEG-based encryption methods are classified into three categories according to the integration mode of encryption modules: encryption-then-compression (ETC), compression-then-encryption (CTE), and joint image compression and encryption (JICE). ETC first encrypts the image, typically using pixel scrambling and diffusion techniques, and then compresses data [12], [13], [14]. However, modifying pixels will destroy the correlation, thereby leading to the compression performance degradation. CTE encrypts the JPEG files postcompression, ensuring minimal impact on compression efficiency [15]. However, encrypting JPEG files often corrupts the format, making it incompatible with standard JPEG workflows. JICE integrate encryption module directly into the compression process, including encrypting transformation coefficients in the transformation stage [16], [17], [18], [19], encrypting quantization tables in the quantization stage [20], and encrypting coefficient bitstreams in the entropy coding stage [21]. JICE can achieve a better balance between efficiency and security without compromising compatibility.

JPEG-based recompression methods have made gratifying progress recently. Itier et al. [22] first tried to design a recompression scheme for JPEG images without decryption and successfully realized the lossy recompression of encrypted JPEG images without the key. Sun et al. [23] designed an LLJPEG lossless recompression scheme based on transform domain intraprediction. It can save about 25%–30% extra storage on the basis of JPEG without introducing any additional information loss. However, these methods are fundamentally constrained by a three-band (RGB) architecture, rendering them suboptimal for multispectral RS data in which removing spatial and spectral correlation plays a critical role for optimal compression.

In this article, we present a novel JECA for multispectral images (JECA-MSs). Unlike existing JPEG-based methods, which are primarily optimized for general digital images, the JECA-MS addresses the unique challenges of Multispectra RS data as follows.

1) *Adaptive Two-Size TBD:* Leveraging differences in texture intensity to segment images into weak texture blocks (WTBs) and strong texture blocks (STBs), optimizing compression efficiency without sacrificing important details.

2) *Modified 3-D Discrete Cosine Transform:* Combining spatial and spectral decorrelation, the decorrelation effect of weak texture areas such as water and clouds is better than that of traditional 3D-DCT.

3) *Ciphertext Domain Recompression:* The encrypted coefficients are scaled by a user-defined recompression factor to realize multiple rounds of recompression in the ciphertext domain, providing a safe and flexible way to adjust the CR for JECA-MS coded data.

4) *JECA-MS Coding Format:* A custom file structure optimized for RS data, reducing overhead compared to traditional JPEG file interchange format (JFIF) [24]. It is the only file organization method suitable for JECA-MS coded data at present.

The outline of the article is as follows. Section II reviews the related work. Section III describes the JECA-MS in detail, including the overall structure, adaptive two-size texture block decision (TBD), encryption, and initial compression, recompression, and decoding. Sections IV and V provide experimental results and conclusions, respectively.

II. Related Works

A. JPEG-Based Encryption Algorithms

JPEG-based encryption algorithms aim to integrate encryption processes into the JPEG compression workflow, balancing security and compression efficiency. Kurihara et al. [13] designed a scheme that is friendly to JPEG compression. The input image is divided into subblocks with a size of 16×16 and the order is scrambled. Then, each subblock is rotated, inverted, negative-positive transformed, and color component shuffled in turn. Finally, subblocks are integrated together to obtain the encrypted image. On this basis, Chuman et al. [14] further designed a scheme with the grayscale image as the object. First, the color space of the image is transformed into YCrCb and combined into a grayscale image. The grayscale image will be divided into subblocks of size 8×8 and scrambled. Then, each subblock will be rotated, inverted, and negative-positive transformed in turn. The blocks divided by a grayscale-based scheme are smaller in size, more in number, less in color information, and larger in key space, so they can obtain higher security. Li et al. [17] adopted a two-level block segmentation strategy to segment the image into blocks with sizes of 8×8 and 16×16 with minimum file size increment and block distortion. The blocks of different sizes are decorrelated by the DCT of corresponding order, and transformed coefficients are encrypted by the DC XOR and dc-excluded block permutation. However, these schemes are RGB digital

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image designs. Their performance will be constrained when processing RS images due to compatibility issues.

B. Recompression of JPEG Images

Itier et al. [22] tried to design a scheme that can directly recompress encrypted JPEG images without key decryption. In the initial compression process, they use XOR to encrypt the transform coefficients, which can keep the bit length unchanged. Then, in the recompression stage, the last bit of the coefficients is removed to realize the recompression in the ciphertext domain. Since the bit length of the coefficients is unchanged before and after encryption, it can be decrypted and decoded normally. This method not only achieves high-security recompression without destroying the format, but also has compression performance close to JPEG. Sun et al. proposed LLJPEG, a lossless recompression scheme for JPEG-coded images based on intraprediction in the transform domain [23]. This approach derives reference pixels by decoding the JPEG image, uses these references for intraprediction to generate prediction blocks, and then applies transformation and quantization to obtain predicted coefficients. The residuals, calculated as the difference between the predicted and original coefficients, are entropy-coded to produce the recompressed bitstream. However, the performance of LLJPEG is heavily dependent on the correlation between original blocks and reference pixels. When LLJPEG is used to recompress encrypted JPEG images, the performance will be significantly reduced due to the destruction of correlation.

III. Proposed JECA-MS

A. Motivation of 3D-MDCT

To enhance compression efficiency, the design of the JECA-MS fully considers the unique characteristics of multispectral RS data, such as uneven texture distribution and strong spatial–spectral correlation. For texture variability, an adaptive two-size TBD mechanism is introduced to divide strong and WTBs. For better spatial–spectral decorrelation, we propose a novel transform, the modified 3-D discrete cosine transform (3D-MDCT), which improves upon traditional 3D-DCT.

Previous studies have employed 3D-DCT for hyperspectral image decorrelation [25], [26], [27]. Building on this foundation, we observe that DCT-VI, a variant of DCT, demonstrates superior spectral decorrelation, particularly for weak-textured images like water and clouds. Accordingly, we design 3D-MDCT, which incorporates DCT-VI into the transformation process. The forward and inverse transformations are defined as follows.

Forward Transformation

$$F(u,v,w)=\frac{4}{N\sqrt{2M-1}}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1}\sum_{w=0}^{M-1}f(x,y,z)\,\varepsilon_u\varepsilon_v\varepsilon_w\eta_z C_{2N}^{u(2x+1)}C_{2N}^{v(2y+1)}C_{2M-1}^{w(2z+1)}. \quad (1)$$

Inverse Transformation

$$f(x,y,z)=\frac{4}{N\sqrt{2M-1}}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1}\sum_{w=0}^{M-1}F(u,v,w)\,\varepsilon_x\varepsilon_y\varepsilon_z\eta_w C_{2N}^{x(2u+1)}C_{2N}^{y(2v+1)}C_{2M-1}^{z(2w+1)} \quad (2)$$

TABLE I
Low-Frequency Energy (P_{lf})

Transform	Water	Cloud	Farmland	Building
3D-DCT	0.87	0.96	**0.98**	**0.97**
3D-MDCT	**0.93**	**0.99**	0.97	0.95

TABLE II
High-Frequency Energy (P_{hf})

Transform	Water	Cloud	Farmland	Building
3D-DCT	0.07	0.14	0.26	0.58
3D-MDCT	**0.06**	**0.12**	**0.23**	**0.51**

where $f(x,y,z)$ represents the original pixel values, $F(u,v,w)$ represents the transformed coefficient, N is 8 or 16, and M is the number of bands. Parameters ε_u and η_z are defined as

$$\varepsilon_u=\begin{cases}\frac{1}{\sqrt{2}}, & u=0\\ 1, & u\neq 0\end{cases},\quad \eta_z=\begin{cases}\frac{1}{\sqrt{2}}, & z=M-1\\ 1, & z\neq M-1.\end{cases} \quad (3)$$

C_j^i is defined as

$$C_j^i=\cos\left(\frac{\pi i}{j}\right). \quad (4)$$

To validate the effectiveness of 3D-MDCT, we analyze the energy distribution across low and high frequencies for weak- and strong-textured images. Assuming the input image is divided into $16\times 16\times M$ blocks, the energy proportions are calculated as follows.

Total Energy

$$E_t=\sum_{u=0}^{16}\sum_{v=0}^{16}\sum_{w=0}^{M-1}|B_{u,v,w}|^2. \quad (5)$$

Low-Frequency Energy

$$E_{lf}=\sum_{u=0}^{7}\sum_{v=0}^{7}\sum_{w=0}^{\frac{M}{2}-1}|B_{u,v,w}|^2. \quad (6)$$

High-Frequency Energy

$$E_{hf}=\sum_{u=8}^{16}\sum_{v=8}^{16}\sum_{w=\frac{M}{2}}^{M-1}|B_{u,v,w}|^2. \quad (7)$$

Energy Proportions

$$P_{\mathrm{lf}}=\frac{E_{\mathrm{lf}}}{E_t},\quad P_{\mathrm{hf}}=\frac{E_{\mathrm{hf}}}{E_t}\times 10^3 \quad (8)$$

where P_{lf} represents the proportion of low-frequency energy (higher values indicate better decorrelation) and P_{hf} means high-frequency energy (lower values indicate better decorrelation). Tables I and II summarize experimental results, which

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demonstrate that 3D-MDCT achieves superior decorrelation, particularly for weak-textured images like water and clouds, compared to 3D-DCT.

B. Overall Framework of the JECA-MS Encoder

The JECA-MS encoder is depicted in Fig. 1. The encoding steps are as follows.

1) *Preprocessing:* The input image will be divided into WTBs and STBs according to the texture intensity by adaptive two-size TBD.

2) *Encryption and Initial Compression:* WTBs and STBs undergo different compression strategies, transformed into the ciphertext domain via block scrambling and DC XOR diffusion. The results are encoded using JPEG entropy coding to produce crypto-compressed bitstreams, which are packaged into files in JECA-MS coding format (JECA-MS-CF) for storage and transmission.

3) *Recompression:* In the ciphertext domain, recompression adjusts CRs by scaling coefficients with a recompression factor, enabling multiple rounds of recompression without decryption.

C. Adaptive Two-Size TBD

To efficiently encode multispectral RS images, the input image is segmented into WTBs and STBs. This segmentation optimizes compression performance by preserving critical texture details in STBs and achieving high compression in WTBs. The process is shown in Algorithm 1 and summarized as follows.

1) *Initial Block Division:* The input image ($H\times W\times 4$) is divided into N blocks of size $16 \times 16 \times 4$, denoted as $BS_{16} = \{B_1^{16}, B_2^{16}, B_3^{16}, \ldots, B_N^{16}\}$, where $N = (H\times W) / (16 \times 16)$.

2) *Texture Analysis with 3D-MDCT:* For each B_i^{16}, perform 3D-MDCT and calculate the sum of absolute ac coefficients, denoted as $SOA_{16}(i)$. Then, divide B_i^{16} into four subblocks of size $8 \times 8 \times 4$, denoted as $B_{\text{sub}}(i) = \{B_i^8, B_i^8, B_i^8, B_i^8\}$, and calculate the sum of absolute ac coefficients $SOA_8(i)$ for each $B_{\text{sub}}(i)$.

3) *Fast TBD:* Compare $SOA_{16}(i)$ and $SOA_8(i)$ by the following formula:

$$d(i) = \text{SOA}_8(i) \times k - \text{SOA}_{16}(i) \tag{9}$$

$$s_i = \begin{cases} 0, & \text{if } d(i) < 0 \\ 1, & \text{if } d(i) \geq 0 \end{cases} \tag{10}$$

where s_i is the decision result and will be put into the decision set S. If s_i is 0, B_i^{16} is classified as STB and then divide into $B_{\text{sub}}(i)$ to preserve texture details. In contrast, if s_i is 1, B_i^{16} will be classified as WTB and its size will be reserved. The control parameter k is used to adjust the proportion of WTBs and STBs, which is crucial in balancing compression efficiency and image quality. Higher k will increase the proportion of WTBs, improving the speed and efficiency of compression but potentially losing some texture details. Lower k will increase the proportion of STBs and keep more details at the expense of reducing compression efficiency.

Algorithm 1 Adaptive Two-Size TBD

Input: Input image ($H \times W \times 4$), control parameter k;
Output: Block set $BS_{8\text{-}16}$ (containing STBs and WTBs), Decision set S (containing 0 and 1);

```
Divide image into N16 × 16 × 4 blocks BS16;
for each block B_i^16 in BS16 do
    Perform 3D-MDCT and calculate SOA16(i);
    Divide B_i^16 into four subblocks B_sub(i);
    Perform 3D-MDCT on B_sub(i) and calculate SOA8(i);
    if SOA8(i) × k < SOA16(i) then
        Classify as STB, add B_sub(i) to BS8-16 and 0 to S;
    else
        Classify as WTB, add B_i^16 to BS8-16 and 1 to S;
    end if
end for
Return BS8-16 and S
```

D. Encryption and Initial Compression

Following the adaptive two-size TBD, the input image is divided into STBs and WTBs, stored in $BS_{8\text{-}16}$ with the corresponding decision set S. In the encryption and initial compression stage, STBs and WTBs are processed using distinct compression strategies to balance texture preservation and compression efficiency. Then, the obtained compressed data are uniformly encrypted and entropy encoded into crypto-compressed bitstreams. The workflow is shown in Fig. 2. STBs aim to preserve critical texture details. Each STB is quantized after being transformed by 3D-MDCT. Let the transformed coefficients are denoted as $\text{DCT}_s(x, y, z)$, the quantization formula is as follows:

$$q_s(x, y, z) = \lfloor \frac{\text{DCT}_s(x, y, z)}{2} \rfloor \tag{11}$$

where $q_s(x, y, z)$ is the quantized coefficient. The divisor prevents coefficient overflow of subsequent entropy coding.

WTBs improve coding speed and compression efficiency through high-frequency zeroing quantization. Let transformed coefficients after 3D-MDCT be defined as $\text{DCT}_w(x, y, z)$, $\text{DCT}_w(0, 0, 0)$ is the dc coefficient, and the rest are ac coefficients. High-frequency zeroing quantization is as follows:

$$q_w(x, y, z) = \begin{cases} \lfloor \frac{\text{DCT}_w(x, y, z)}{4} \rfloor, & \text{if } (x, y, z) = (0, 0, 0) \\ \lfloor \frac{\text{DCT}_w(x, y, z)}{2} \rfloor, & \text{otherwise} \end{cases} \tag{12}$$

$$q_h(x, y, z) = \begin{cases} q_w(x, y, z), & \text{if } x, y \in [0, 8],\ z \in [0, 4] \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where is the quantized coefficient of WTB. The dc and ac coefficients will be divided by 4 and 2, respectively, to prevent overflow. Fig. 3 specifically demonstrates 3D-MDCT and high-frequency zeroing quantization of WTB. The zeroed quantized coefficients are discarded, each WTB is cut to 8

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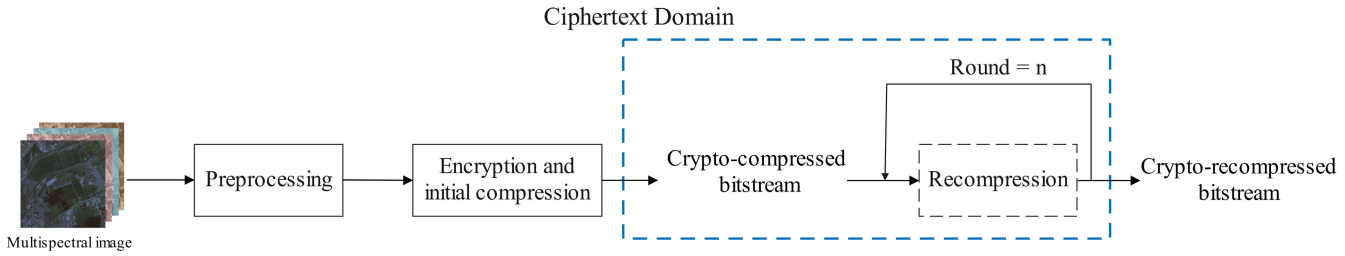


Fig. 1. Overall structure of the JECA-MS encoder.

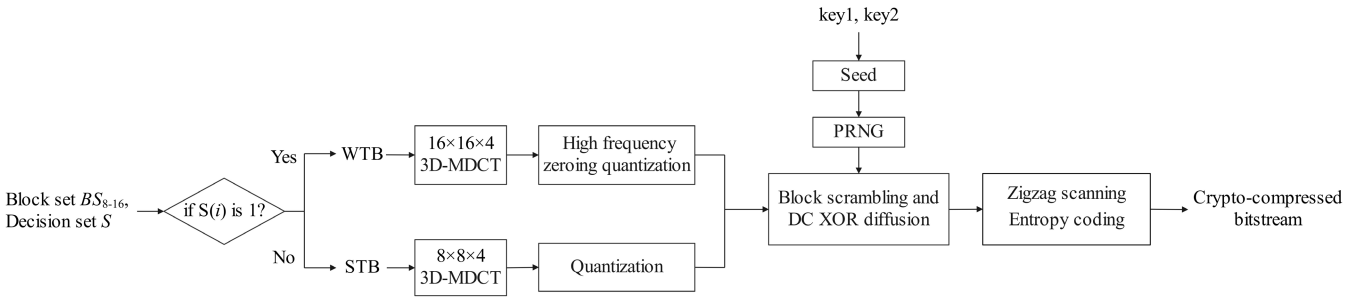


Fig. 2. Process of initial compression and encryption.

$\times$ 8 $\times$ 4 size. The quantized STBs and WTBs will be used as the minimum coding units (MCUs) for ciphertext domain conversion, zigzag scanning, and entropy coding.

The ciphertext domain conversion of MCUs can be realized by block scrambling and DC XOR diffusion with a pseudorandom number generator (PRNG), plain-image-dependent key generation mechanism [17], or chaotic mapping [19]. Take PRNG as an example, key1 and key2 are used as the input seed of PRNG to obtain unordered index vectors UIV1 and UIV2. The UIV1 is used to scramble the arrangement of MCUs. Note that the seed will change in each band of MCUs to produce different UIV1, that is, if the input is a four-band image, four different UIV1 will be generated to scramble each band of MCUs, respectively. The UIV2 is used to perform a bitwise XOR operation, excluding the sign bits, on the dc coefficients in each MCU. Assuming that the bit length of the dc coefficient is L, the DC XOR diffusion is as follows:

$$E_{\mathrm{dc}}(i) = M_{\mathrm{dc}}(i) \oplus \mathrm{UIV2}(i), \quad i = 0, 1, 2, \ldots, L \tag{14}$$

where E_{dc} is the dc coefficient after XOR diffusion, M_{dc} is the original dc coefficient, and UIV2 is a PRNG-generated sequence dependent on key2 and L. Each MCU will generate a unique UIV2 for diffusion.

Since the size of each band of MCU is 8 $\times$ 8, which is the same as that of JPEG, it can also adopt the zigzag scanning and entropy coding provided by JPEG to obtain crypto-compressed bitstreams. Each band of the MCU is zigzag scanned into vectors and then entropy coded. The decision set $S(i)$ will be embedded in the header of each MCU. For the encrypted dc coefficient, the difference with the encrypted dc coefficient of the previous adjacent MCU is calculated, called the prediction error. Then, the Huffman algorithm is performed on the prediction error to obtain the crypto-compressed bitstream.

TABLE III
FILE STRUCTURE OF JECA-MS-CF

Name	Marker	Byte length	Description
SOI	0xFFD8	none	Start Of Image
APP0	0xFFE0	16	Identification Code
SOF0	0xFFC0	13	Start Of Frame
DHT	0xFFC4	31	Huffman Tables
SOS	0xFFDA	variable	Start Of Scan
Crypto-compressed bitstreams			
EOI	0xFFD9	none	End Of Image

For ac coefficients, run-length coding is used to encode continuous zero values and perform the Huffman algorithm together with nonzero values to obtain the crypto-compressed bitstream.

E. JECA-MS-CF Analysis

The crypto-compressed bitstream is encapsulated by JECA-MS-CF for storage and transmission. The overall structure of JECA-MS-CF is shown in Table III.

Compared with JFIF [24], JECA-MS-CF employs a simplified header structure, reducing the bit overhead, which is particularly beneficial for small-sized images. The key modifications include as following.

1) *Removal of DQT Field:* There is no need to use quantization tables for the JECA-MS.

2) *SOF0 Adjustments:* Add the extra 6-bit spaces to store band, number of blocks, recompression factor, and recompression times. The modified SOF0 field is shown in Table IV.

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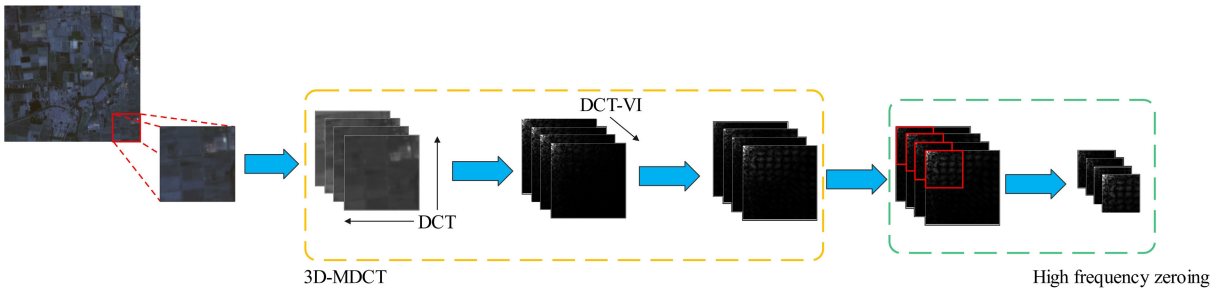


Fig. 3. Perform 3D-MDCT and high-frequency zeroing on WTBs.

TABLE IV
SOF0 FIELD STRUCTURE OF JECA-MS-CF

Name	Byte length	Value
Marker	none	0xFFC0
Length	2	0x000D
Sample precise	1	0x08
Height	2	0-0xFFFF
Width	2	0-0xFFFF
Band	2	0-0xFFFF
Number of blocks	2	0-0xFFFF
Recompression factor	1	0-0xFF
Recompression times	1	0-0xFF

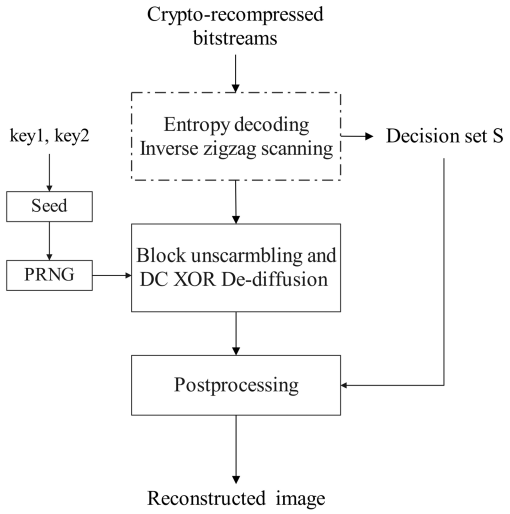


Fig. 4. Process of the decoding phase.

F. Recompression

The JECA-MS enables repeated recompression directly in the ciphertext domain, thanks to the unchanged bit length of encrypted coefficients. The recompression workflow is as follows.

1) Decode the SOS field and perform entropy decoding to extract dc coefficients C_{dc} and ac coefficients C_{ac}.

2) Achieve recompression by dividing coefficients by the recompression factor r extracted from the SOF0 field. The formula is as follows:

$$C'_{\mathrm{dc,ac}} = \lfloor \frac{C_{\mathrm{dc,ac}}}{r} \rfloor \tag{15}$$

where C'_{dc} and C'_{ac} are recompressed coefficients and $\lfloor * \rfloor$ represents rounding down. The default value of r is 2.

3) Re-entropy encode recompressed coefficients and update the SOS field. The recompression times in the SOF0 field increment by 1.

This process does not modify coding format, allowing for multiple recompression cycles.

G. Decoding Phase

The purpose of decoding is to decrypt JECA-MS coded data and reconstruct it into the image. It is in reverse order to the encoding, as shown in Fig. 4. The specific process of decoding is as follows.

1) The crypto-recompressed bitstreams will be entropy decoding and inverse zigzag scanning first to get encrypted MCUs, $S(i)$ embedded in the header of each MCU is extracted in turn to form the decision set S.

2) Key1 and key2 are input into PRNG as a seed to obtain UIV1 and UIV2, which are the same as those generated by the encoder. The MCUs are grouped by bands, and the corresponding UIV1 is used for block unscrambling. For the encrypted dc coefficients in each band of MCUs, the XOR operation of (14) is used again for de-diffusion. The restored MCUs are stacked in turn to obtain $BS_{8\text{-}16}$.

3) Postprocessing is performed on $BS_{8\text{-}16}$ to get WTBs and STBs. The sub-WTBs and sub-STBs are extracted from $BS_{8\text{-}16}$ according to decision set S. If $S(i) = 1$, it needs to extract one sub-WTB from $BS_{8\text{-}16}$ and build three $8 \times 8 \times 4$ size zero blocks to splice into one WTB, where zero blocks are generated by the decoder; if $S(i) = 0$, it needs four sub-STBs are extracted from $BS_{8\text{-}16}$ and then spliced into one STB. For WTBs, inverse quantization and (2) are performed on each WTB. For STBs, inverse quantization and (2) are performed on four sub-STBs before splicing into one STB. The inverse quantization formula is as follows:

$$q'_h(x,y,z) = \begin{cases} q_h(x,y,z) \times 4 \times r^n, & \text{if } (x,y,z) = (0,0,0) \\ q_h(x,y,z) \times 2 \times r^n, & \text{otherwise} \end{cases} \tag{16}$$

$$q'_s(x,y,z) = q_s(x,y,z) \times 2 \times r^n \tag{17}$$

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TABLE V
COMPARISON OF ENCODING TIME (IN SECONDS)

Methods	Water	Cloud	Farmland	Building	Average
Chuman et al. [14]	11.13	12.67	22.56	42.81	22.29
Qin et al. [16]	30.69	38.41	42.70	62.39	43.55
Li et al. [17]	14.04	15.65	24.70	40.64	23.76
Yuan et al. [18]	34.71	36.69	41.12	65.11	44.41
JECA [19]	12.65	26.70	31.98	41.45	28.20
Itier et al. [22]	24.81	40.02	50.75	62.26	44.46
LLJPEG [23]	18.61	20.29	43.88	45.98	32.19
JECA-MS	**10.72**	**10.67**	**21.67**	**39.83**	**20.72**

TABLE VI
COMPARISON OF DECODING TIME (IN SECONDS)

Methods	Water	Cloud	Farmland	Building	Average
Chuman et al. [14]	5.73	5.85	7.51	8.68	6.94
Qin et al. [16]	19.09	21.99	24.24	31.51	24.21
Li et al. [17]	4.43	5.01	5.97	7.92	5.83
Yuan et al. [18]	18.91	20.33	21.83	27.41	22.12
JECA [19]	7.17	11.34	12.39	28.33	14.81
Itier et al. [22]	5.59	7.64	13.20	17.37	10.95
LLJPEG [23]	5.04	5.18	9.37	10.25	7.46
JECA-MS	**2.11**	**2.48**	**4.09**	**8.23**	**4.23**

TABLE VII
AVERAGE PSNR, OA, AND KAPPA UNDER DIFFERENT K VALUES (CR $\in$[60, 120])

k	PSNR	OA	Kappa
0.7	38.91	0.9898	0.9686
0.8	40.83	0.9906	0.9714
0.9	41.89	0.9909	0.9728
1.0	**42.43**	**0.9916**	**0.9746**

where $q'_h(x, y, z)$ and $q'_s(x, y, z)$ are coefficients of WTBs and STBs, r is the recompression factor, and n is the recompression times.

4) According to the width, height, and bands obtained by the SOF0 field, WTBs and STBs are integrated together to reconstruct the image.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

We have established a dataset covering four typical ground objects: water, cloud, farmland, and building images. Water and cloud images are from Tiangong-2. Its original data consists of 14 channels, of which 1–5 are near red bands, 6–9 are red bands, 10–11 are green bands, and 12–14 are blue bands. We select one channel from each band to form a four-channel image. Farmland and building images are from Gaofen-2, which consists of 4 channels: blue, green, red, and near red. Each image is an 8-bit pseudo-color image with the size of 1024 × 1024 × 4.

In the following experiment, the methods compared with the JECA-MS include JPEG encryption algorithm [14], [16], [17], [18], [19], the most advanced JPEG ciphertext domain recompression algorithm [22], and the latest JPEG lossless (LLJPEG) recompression algorithm [23], among which LLJPEG is excluded from the security analysis as it does not implement a new encryption module. The experiments are conducted on Python 3.9, the processor is Intel[1] Core[2] i7-14700HX 2.10 GHz, and the graphics card is NVIDIA GeForce RTX 4070.

B. Time Complexity

The encoding and decoding times are tested to compare the complexity of the JECA-MS with other methods. The results are shown in Tables V and VI. Compared with similar methods, the JECA-MS achieves average speed improvements of 39% in encoding and 68% in decoding, significantly reducing computational overhead while enhancing processing

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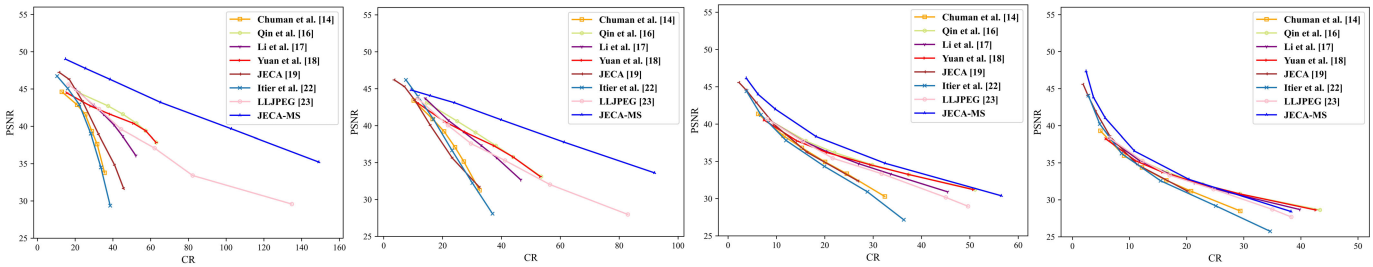


Fig. 5. Comparison of compression performance with different methods. (a) Water, (b) cloud, (c) farmland, and (d) building.

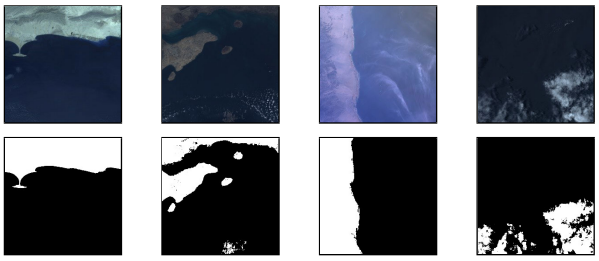


Fig. 6. Original images and corresponding reference mask images.

efficiency. These advantages make the JECA-MS suitable for resource-constrained or real-time-critical applications. The speed improvement of the JECA-MS mainly benefits from fast TBD and high-frequency information discarding of WTBs. Fast TBD minimizes decision time while ensuring accurate division of WTBs and STBs. Moreover, WTBs are processed with simplified encoding by discarding three-quarters of high-frequency information, significantly reducing the amount of computation.

C. Compression Performance

The compression performance is tested by calculating the peak signal-to-noise ratio (PSNR) under different CRs. The PSNR can quantitatively describe the difference between the reconstructed and the original image. The larger the value, the higher the reconstructed image quality. CRs can be calculated as follows:

$$\mathrm{CR} = \frac{\text{original image file size}}{\text{compressed image file size}} \times 100\%. \tag{18}$$

The drawn CR-PSNR curve is as shown in Fig. 5. LLJPEG uses the same encryption scheme as the JECA-MS to process the input JPEG images. It can be seen that the JECA-MS is obviously superior to other methods in dealing with water and cloud images. With similar PSNR, the CR is increased by 55% and 36%, respectively, compared with the most advanced methods. This is because the texture details of water and clouds are few, enabling the JECA-MS to achieve high compression while ensuring image quality. For farmland and building images with strong textures, although the performance is reduced, the JECA-MS still has advantages at low CR.

D. Effect Comparison on Water Extraction

This part takes water extraction as an example to test the effect of making RS products with JECA-MS-coded images. We use the normalized difference water index (NDWI) [28] to extract water pixels. For the quantitative description, overall classification accuracy (OA) and the Kappa coefficient are used. Their formulas are defined as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{19}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{20}$$

$$P_e = \frac{(\mathrm{TP} + \mathrm{FP}) \times (\mathrm{TP} + \mathrm{FN}) + (\mathrm{TN} + \mathrm{FP}) \times (\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} \tag{21}$$

where TP is the number of pixels in both the reference mask image and the compressed mask image that are water bodies. TN represents the nonwater body parts in both the reference mask image and the compressed mask image. FP indicates the part that is a water body in the compressed mask image but not in the reference mask image. FN is the opposite of FP.

The experiment is designed as follows.

1) Calculating NDWI for uncompressed water images, and the reference mask images are obtained by masking.
2) Calculating NDWI and masking water images encoded by the JECA- MS to obtain compressed mask images.
3) OA and Kappa coefficients between the reference mask image and the compressed mask image are calculated.

Examples of the obtained reference mask image are shown in Fig. 6, in which the black areas are water bodies and the white areas are nonwater bodies. Note that for some thin clouds that can be obviously inferred as water below, we mask them as water bodies.

Fig. 7 shows the results of OA and Kappa coefficients. The effect of the JECA-MS is still superior to similar schemes. In addition, we can find that the accuracy of Chuman et al. [14] scheme obviously increases in the process of CR from 20 to 30, which is different from the PSNR values. This is because high-frequency information contains noise besides texture details, and the increase of CR will remove some high-frequency information, which has the effect of denoising. Although lossy compression inevitably discards certain image details, it simultaneously removes noise, thereby enhancing the robustness and accuracy of downstream RS applications.

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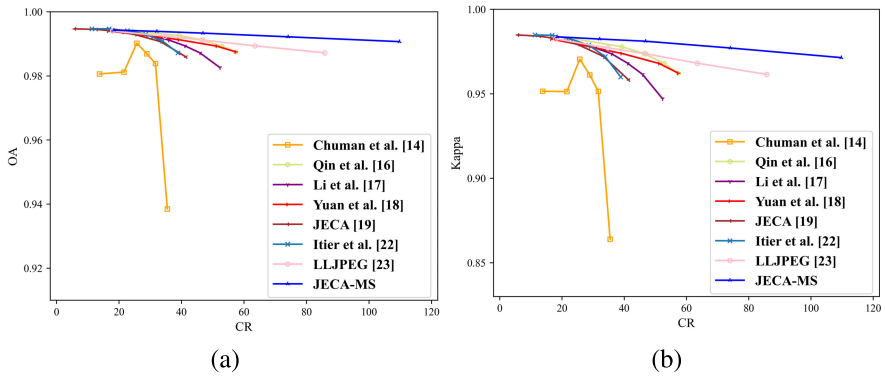


Fig. 7. Comparison of OA and kappa with different methods. (a) CR-OA curve and (b) CR-Kappa curve.

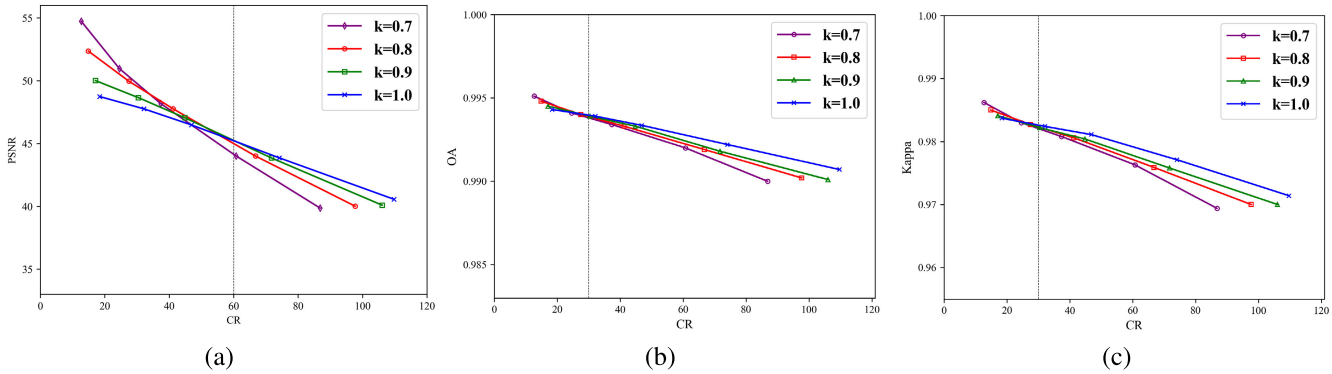


Fig. 8. Comparison of PSNR, OA, and kappa of the JECA-MS with k values of 0.7, 0.8, 0.9, and 1.0. (a) CR–PSNR curve, (b) CR–OA curve, and (c) CR–kappa curve.

TABLE VIII
MSE OF ENCRYPTED MASK IMAGES UNDER DIFFERENT CRS

Methods	20	30	40	50	Average
Chuman et al. [14]	0.5430	0.5432	0.5439	0.5435	0.5434
Qin et al. [16]	0.6307	0.6298	0.6295	0.6292	0.6298
Li et al. [17]	0.5855	0.5869	0.5957	0.6280	0.5990
Yuan et al. [18]	0.6266	0.6291	0.6292	0.6285	0.6284
JECA [19]	0.5483	0.5426	0.5526	0.5573	0.5502
Itier et al. [22]	0.4896	0.4897	0.4823	0.4854	0.4868
JECA-MS	**0.6398**	**0.6395**	**0.6390**	**0.6381**	**0.6391**

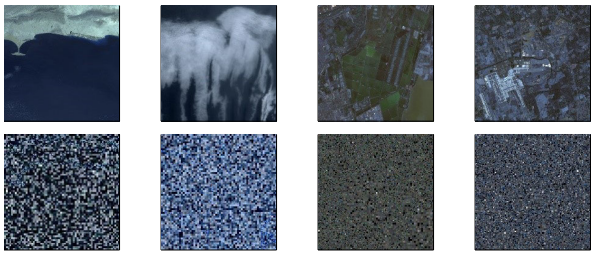


Fig. 9. Visual effect of encrypted JECA-MS-coded images.

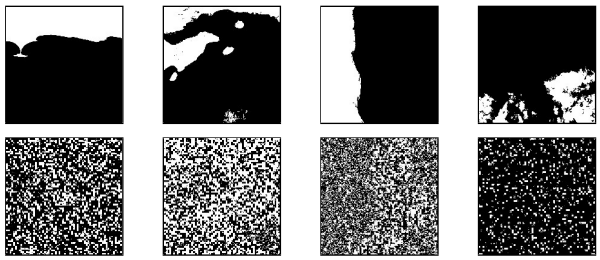


Fig. 10. Visual effect of compressed mask images obtained by encrypted JECA-MS-coded images.

The relationship between lossy compression and denoising has been widely studied in the fields of signal and image processing [29], [30], [31]. In the RS field, there are also some studies to explore the relationship between lossy compression and statistical parameter retrieval [32].

The JECA-MS removes high-frequency information by high-frequency zeroing on WTBs, while STBs keep high-frequency information as much as possible. The balance between denoising and texture detail preservation of the JECA-MS can be realized by adjusting k values to control the proportion of WTBs and STBs. The k value sensitivity analysis

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TABLE IX
INFORMATION ENTROPY OF ENCRYPTED WATER (W) AND CLOUD (C) IMAGES UNDER DIFFERENT CRS

Methods		20	30	40	50	Average
Chuman et al. [14]	W	7.8566	7.8265	7.6828	7.5552	7.7303
	C	7.5971	7.5564	7.5168	7.4473	7.5294
Qin et al. [16]	W	7.8268	7.7620	7.6924	7.6673	7.7371
	C	7.5856	7.5807	7.5717	7.5487	7.5717
Li et al. [17]	W	7.6740	7.6566	7.6470	7.6451	7.6557
	C	7.6070	7.6039	7.5960	7.5814	7.5971
Yuan et al. [18]	W	7.8634	7.8553	7.8454	7.8180	7.8455
	C	7.6072	7.6068	7.6056	7.6023	7.6055
JECA [19]	W	7.0289	7.0391	7.0422	7.0041	7.0286
	C	7.0346	7.0433	7.0442	7.0065	7.0322
Itier et al. [22]	W	7.8514	7.8479	7.8327	7.7460	7.8195
	C	7.6087	7.5967	7.5164	6.7861	7.3770
JECA-MS	W	**7.8658**	**7.8652**	**7.8636**	**7.8607**	**7.8638**
	C	**7.6124**	**7.6110**	**7.6107**	**7.6097**	**7.6110**

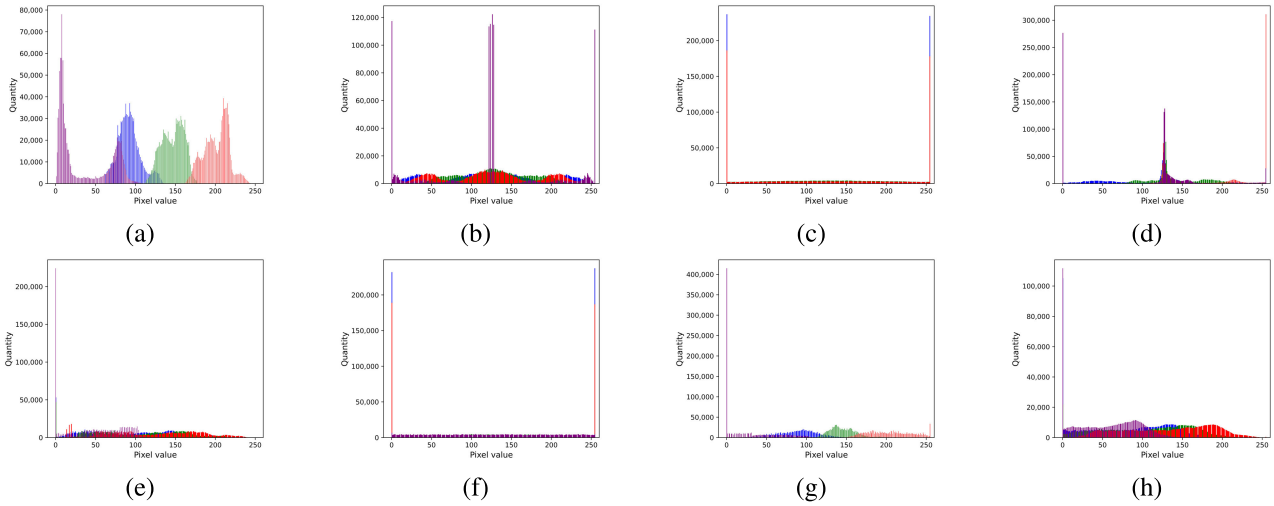


Fig. 11. Histogram distribution of different methods. (a) Origin image. (b) Chuman et al. [14]. (c) Qin et al. [16]. (d) Li et al. [17]. (e) Yuan et al. [18]. (f) JECA [19]. (g) Itier et al. [22]. (h) JECA-MS.

TABLE X
PERFORMANCE TEST OF THE JECA-MS WITH TBD OR NOT (NO-TBD)

Indices		1	2	3	4	Average
Time	TBD	**10.67**	**12.35**	**13.35**	**13.98**	**12.59**
	NO-TBD	10.72	15.84	17.71	19.28	15.89
CR	TBD	**25.61**	**38.60**	**65.39**	**102.86**	**58.12**
	NO-TBD	15.79	23.00	34.45	45.21	29.61
PSNR	TBD	47.75	46.28	**43.19**	**39.65**	44.22
	NO-TBD	**50.36**	**47.21**	42.75	38.24	**44.64**
Kappa	TBD	**0.9827**	**0.9793**	**0.9724**	**0.9157**	**0.9625**
	NO-TBD	0.9818	0.9786	0.9306	0.7728	0.9160

has been conducted, as shown in Fig. 8, where we evaluated the effect of k (values: 0.7, 0.8, 0.9, and 1.0) on CR, image quality (PSNR), and downstream water extraction performance (OA and kappa). It can be seen that when the CR is less than

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about 60, the image quality is inversely proportional to the k values. However, for water extraction performance, unless the CR is low (less than 30), higher k values will always maintain the advantage of accuracy. Keeping high values of the PSNR, OA, and kappa simultaneously can achieve the best balance between texture detail preservation and denoising. When CR∈[60, 120], the average results of the PSNR, OA, and kappa are shown in Table VII. Since RS data tend to be stored and transmitted with high CR (for water images, it generally exceeds 60), we recommend setting k to 1.0 for water-related scenarios. This value divides more areas into WTBs, providing a strong denoising effect (by eliminating noises that appear as weak textures), yet retains sufficient spectral–spatial consistency to support downstream analysis.

E. Security Analysis

In this part, the security of the JECA-MS is tested, including visual security, RS application security, and the ability to resist statistical attacks.

1) Visual Security: We decode the JECA-MS-coded images without decryption to obtain encrypted images. The visual effects of four typical ground objects, encrypted images are tested: water, cloud, farmland, and building. Original images and their corresponding encrypted images coded by the JECA-MS are shown in Fig. 9. It can be seen that it is hard to get any information from the encrypted images visually.

2) RS Application Security: To test the security of making RS products by JECA-MS-coded images, we use encrypted JECA-MS-coded images to perform the NDWI water extraction experiment. The original reference mask images and compressed mask images obtained by encrypted JECA-MS-coded images are shown in Fig. 10. It can be seen that the compressed mask images are completely different from the original, indicating that the exact position of the water body cannot be obtained from encrypted JECA-MS-coded images. The quantitative test is realized by the mean-square error (mse) between the reference mask image and the encrypted mask image. The formal definition of mse is as follows:

$$\text{mse} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(I_1(i,j) - I_2(i,j))^2 \tag{22}$$

where I_1 and I_2 are the reference mask image and encrypted mask image, respectively. M and N are width and height of mask images, respectively. Since mask images are all binary images, the range of mse is [0,1]. The closer to 1, the higher the security. Under different CRs, the mse of mask images obtained by the JECA-MS and other similar methods is shown in Table VIII. It can be seen that the change of the CR has little influence on mse. The average mse value of the JECA-MS is the highest among similar methods, reaching 0.6391. The above results show that the JECA-MS can ensure the security of making RS products.

3) Resistance to Statistical Attacks: In the statistical attack, the attacker will try to explore the relationship between the original and encrypted image through statistics-based methods. The ability to resist statistical attacks can be evaluated by testing the histogram distribution and information entropy. The histogram distribution is shown in Fig. 11, where the red, green, blue, and purple histograms correspond to bands 1 to 4, respectively. It can be seen that the histogram distribution of the encrypted image is obviously different from that of the original. Affected by compression, a large number of pixels in the encrypted image are concentrated around 0 or 255, while the large change of high-frequency information makes some pixels close to 127, indicating that the encrypted images lose their statistical resemblance. Table IX compares the information entropy of water and cloud images under different CRs; the JECA-MS encrypted images can also be kept at high entropy values. These results prove that the JECA-MS can resist statistical attacks.

F. Ablation Experiment

The ablation experiment is designed to test the effect of TBD or not (NO-TBD) on the performance of the JECA-MS, which is shown in Table X. This experiment is carried out on water images. The test indices include encoding time (time), CR, PSNR, and Kappa. The JECA-MS-coded images will be recompressed from 1 to 4 rounds. It can be seen that for NO-TBD, not only does the encoding time overhead increase by 26% but also the CR decreases by nearly 50%. In fact, NO-TBD is equivalent to all regions being divided into STBs to preserve texture details. However, the texture details in water images are few and contain noise. Preserving them will not only increase the coding cost and data volume but also reduce the accuracy of water extraction, in exchange for only a slight improvement in image quality. These results confirm that dividing strong and WTBs in the JECA-MS significantly enhances compression efficiency and encoding speed while maintaining a reasonable balance of image quality, especially for water images.

V. Conclusion

In this article, a JECA-MS was proposed, including adaptive two-size TBD, 3D-MDCT, a lightweight encryption module (block scrambling + DC XOR diffusion), and a novel JECA-MS-CF. Experimental results demonstrated that the JECA-MS achieved 55% and 36% higher CRs for water and cloud images, compared to state-of-the-art methods, along with 39% faster encoding and 68% faster decoding. It can also maintain the fidelity of downstream tasks and resist statistical attacks.

To our knowledge, the JECA-MS is the first framework to jointly support encryption, compression, and ciphertext domain recompression for multispectral images. The JECA-MS has an inherent balance between compression performance and encryption strength. The current encryption module prioritizes lightweight integration and recompression compatibility, which may not satisfy scenarios requiring cryptographic maximalism. This design choice is intentional—by forgoing complex encryption operations, the JECA-MS maintains the ability to perform lossy recompression directly on ciphertext while achieving real-time processing speeds critical for satellite onboard systems. While this approach may not meet maximalist security requirements, the JECA-MS remains extensible—higher-level encryption modules can be

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incorporated at the cost of compression efficiency, preserving flexibility for diverse scenarios. Future work will explore the adaptation of the dynamic security-compression tradeoff under the JECA-MS framework.

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